

RDKit: State of the Toolkit

2025 UGM edition

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What's new in the last year?

That comes later :-)

First let's talk about the state of the toolkit.

Adoption / usage

Unlike with web apps or commercial software, this is tricky to figure out with open source tools, but let’s try.

Usage: Conda install counts (last release)

Type	Size	Name	Uploaded	Downloads
conda	19.1 MB	linux-64/rdkit-2025.03.5-py312h089baad_0.conda	1 month and 3 days ago	7998
conda	19.3 MB	linux-64/rdkit-2025.03.5-py311hf7fb0bc_0.conda	1 month and 3 days ago	7307
conda	18.9 MB	linux-64/rdkit-2025.03.5-py310h8980e5f_0.conda	1 month and 3 days ago	3995
conda	19.2 MB	linux-64/rdkit-2025.03.5-py313hd383382_0.conda	1 month and 3 days ago	2985
conda	17.8 MB	osx-arm64/rdkit-2025.03.5-py311h1da7121_0.conda	1 month and 3 days ago	2146
conda	17.8 MB	osx-arm64/rdkit-2025.03.5-py312h1ee979d_0.conda	1 month and 3 days ago	1908
conda	18.9 MB	linux-64/rdkit-2025.03.5-py39h7dc6229_0.conda	1 month and 3 days ago	1771
conda	17.8 MB	osx-arm64/rdkit-2025.03.5-py313hf9cf9ea_0.conda	1 month and 3 days ago	999
conda	16.8 MB	win-64/rdkit-2025.03.5-py313h2e2bd31_0.conda	1 month and 3 days ago	637

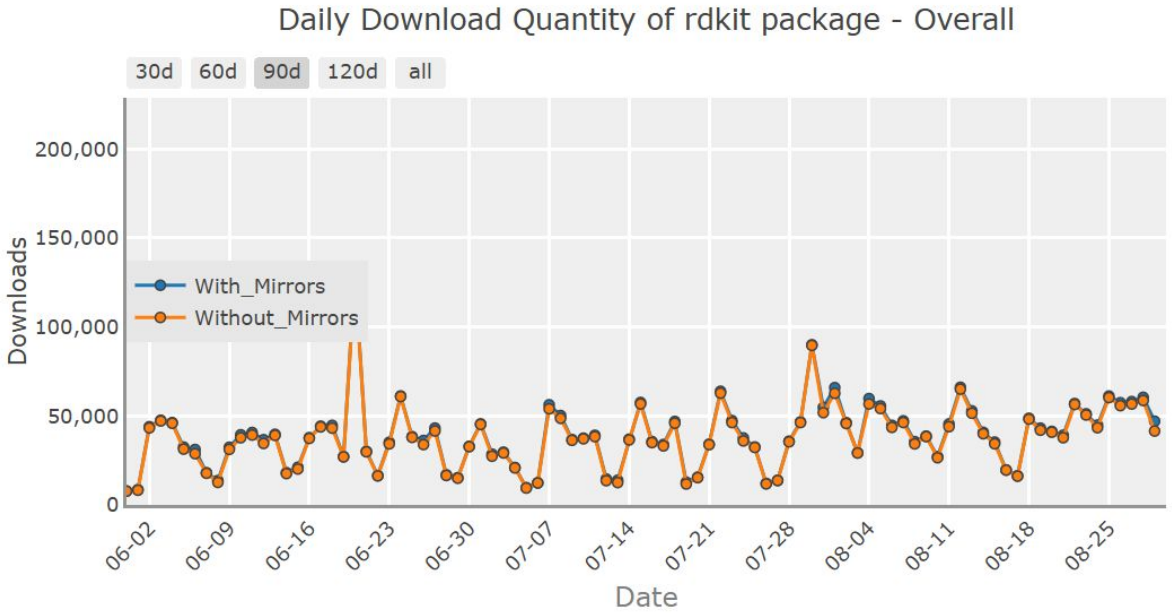
Partial data. Unfortunately the condastats package no longer works

Usage: nuget downloads



Thanks to Gareth Jones for setting this up

Usage: PyPi



Thanks to Chris Kuenneth for getting the pypi installs set up!

Last 90 days of data from https://pypistats.org/packages/rdkit

rdkit-js usage:

rdkit.js

A powerful cheminformatics and molecule rendering toolbelt for JavaScript



Explore the docs »

Repository

github.com/rdkit/rdkit-js

Homepage

www.rdkitjs.com

Weekly Downloads

6’375



Version

2025.3.4-1.0.0

License

BSD-3-Clause

Thanks to Michel Moreau for getting this set up!

Beyond download counts: what about other approaches for looking at adoption?

Usage in online tools/resources

- ChEMBL/SureChEMBL
- ZINC
- Google Patents
- PDBe
- Enamine
- ChemSpace
- TeachOpenCADD
- Probes & Drugs (from CZ-OpenScreen)
- Chemdle

Disclaimer: this info is from public statements made by people associated with those projects. I almost certainly have forgotten someone

Usage in commercial tools

- Amazon Web Services
- Collaborative Drug Discovery
- Cresset Software
- Dalke Scientific Software
- Datagrok
- Glysade
- MedChemica
- NextMove Software
- Schrödinger
- SCM
- Wolfram Research

Disclaimer: this info is from public statements made by people from those companies.
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Github community stats

Community insights

Period: Last year

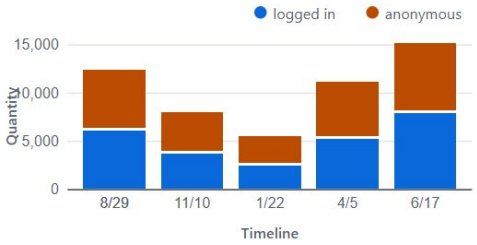
Contribution activity

Count of total contribution activity to Discussions, Issues, and PRs



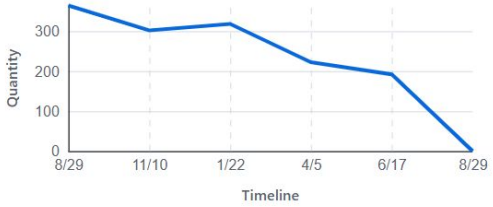
Discussions page views

Total page views to Discussions segmented by logged in vs anonymous users.



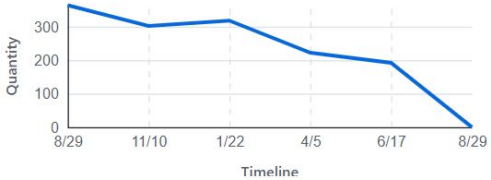
Discussions daily contributors

Count of unique users who have reacted, upvoted, marked an answer, commented, or posted in the selected period.



Discussions new contributors

Count of unique new users to Discussions who have reacted, upvoted, marked an answer, commented, or posted in the selected period.



Other adoption measures

- Mailing lists: 14 messages to rdkit-discuss from 2024.09 - 2025.08[1]
- Google scholar: ~4500 hits for "rdkit" in 2024, >4000 so far in 2025
- Searching github for `"from rdkit import Chem"` returns >75000 code results
- UGM attendance!
- New US UGM!

[1]The mailing list is dying. This is expected.

Sustainability: the bus problem

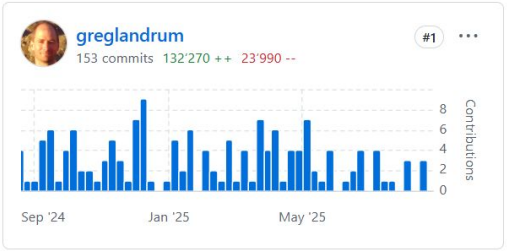
Sustainability: the bus problem

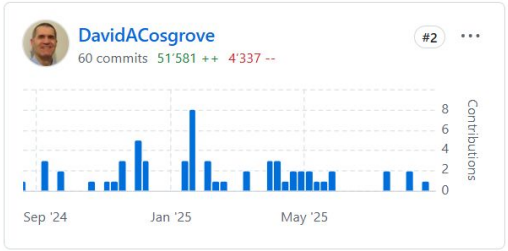
RDKit maintainers:

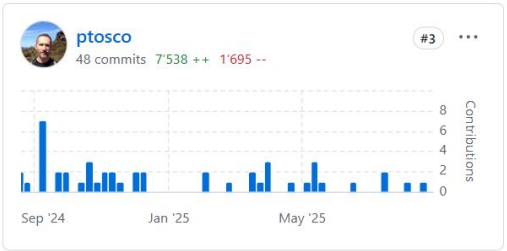
- Greg
- Brian Kelley (Glysade)
- Ricardo Rodriguez Schmidt (Schrödinger)
- Paolo Tosco (Novartis)

Most frequent code contributors in the last year

From 24.08.2024 - 24.08.2025

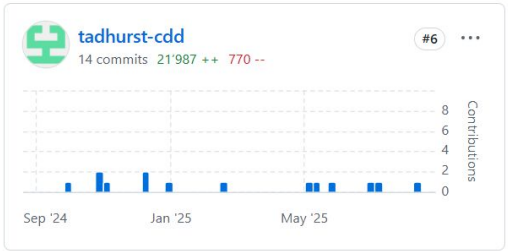












Merged pull request contributors so far in 2025

```
'ptosco', 'Mblakey', 'yishutu', 'KevKeating', 'whosayn',
'iwyoo', 'badisa', 'nrontsis', 'RubenChM',
'rachelnwalker', 'knalice', 'mattwedge-elixir',
'nmaeder', 'e-kwsm', 'richardjgowers', 'pechersky',
'd-b-w', 'rvianello', 'jones-gareth', 'al42and',
'paconius', 'ZontaNicola', 'DavidACosgrove',
'bp-kelley', 'thomp-j', 'IridiumOxide', 'evasnow1992',
'ricrogz', 'scal444', 'greglandrum', 'asiomchen',
'bbu-imdea', 'priley-vv', 'bertiewooster',
'ndickson-nvidia', 'cho-m', 'tadhurst-cdd', 'padix-key'
```

That's 38 different people

Maintenance work in the last year

We started tracking maintenance/cleanup work with the 2019.09 release.

For the 2025.03 and 2025.09 releases, there have been >10 “cleanup” issues/PRs merged:

Roadmap

Future work tends to be determined by what's needed for active projects or requests that come out of the community. So there's not much of a roadmap.

Still, some parts of the way forward are pretty obvious...

✔ ~~Making sure all the pieces required to build a good compound registration system are there~~

Making sure all the pieces required to build a good corporate chemical database are there

Better support for polymers and organometallics

Performance improvements

Ongoing improvements to the conformer generator

Ongoing refactoring and code cleanup

Taking big steps forward…

Some things are hard...

Technology changes (i.e. taking advantage of new C++ or Python versions) is tricky: which operating systems/compilers are people using?

Is it safe to remove old code that seems peripheral or redundant with functionality provided better by other packages?

There are some larger API changes to clean up old mistakes and improve performance and safety that it would be nice to make.

We really, really want to avoid the Python 2/Python 3 situation, so we can’t just make arbitrary changes.

… what we're doing about it

Try to minimize hard external dependencies

Be conservative about language versions/features

Announce deprecations at least one major release in advance

"Backwards incompatible changes" tags

Changing the RDKit release model

Motivation: make new functionality available sooner

Previous:

- Feature releases twice a year, e.g. **2023.03**
  - Possibly including backwards-incompatible changes
- Patch releases every 4-6 weeks, e.g. 2023.03**.2**
  - Only bug fixes, but these can still change results

Current:

- Major releases twice a year, e.g. **2024.03**
  - Possibly including backwards-incompatible changes
- Minor releases every 4-6 weeks, e.g. 2024.03.**2**
  - Include bug fixes (can change results)
  - Include backwards-compatible new features

Ongoing big changes

- Fixing the Hydrogen mess
- Changing the default stereo perception algorithm

More info on these in the What's New notebook

Summary: great stuff

- Steadily growing numbers of people using and building things with the RDKit
- Steady progress on adding new features, fixing bugs, and cleaning up old stuff

Summary: less great

- The group of people making *regular* contributions to the RDKit itself is pretty static
- A steadily growing user community is a mixed blessing: there are a bunch of unanswered (or partially answered) questions in GitHub Discussions

We need:

- More people involved with the code (both C++ and Python) itself
- More people actively helping in the Discussions

https://www.rdkit.org/docs/GettingStartedWithContributing.htm

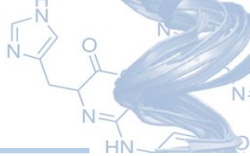
I don't know how to solve this, maybe *we* can figure it out.

How non-developers can contribute:

- Use the code in your own projects and provide feedback:
  - Good bug reports
  - Ideas for improvements
  - Positive feedback via the mailing list/Github discussions
- Answering questions on the mailing list/Github discussions
- Improve the documentation
  - in-code documentation
  - the “Getting started in Python” book
  - the “RDKit Book” reference
  - the “Cookbook”
- Write blog posts (either your own or for the RDKit blog)
- Contribute interesting scripts/libraries for the Contrib folder
- Pay someone else to work on RDKit code[1]

[1] It’s generally a good idea to check with Greg and/or one of the other maintainers before adding significant new functionality.

State of the RDKit?



Ok, enough of that, let's look at what's new

The notebook I'm using will be in the UGM github repo:
https://github.com/rdkit/UGM_2025